ASSIGNMENT · 10

Problem statement: Write a program in C to stimulate a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication & division. The calculator should also perform special operations like computing x^y & $x!$

Objective:
- To understand the use of switch-case statements in C
- To perform basic arithmetic & special mathematical operations.
- To enhance problem-~~stat~~solving skills using menu-driven programs.

Theory: A calculator is a basic computing device used to perform arithmetic operations. In C programming a simple calculator can be implemented using conditional or switch-case statements. Arithmetic operations like addition, subtraction, multiplication & division are fundamental operations.

Platform: Linux

Algorithm:

Step 1: START

Step 2: Declare variables for two numbers x, y, result, & choice.

Step 3: Display menu with operations.
- Addition
- Subtraction
- Multiplication
- Division
- Power (x^y)
- Factorial ($x!$)

Step4: Read the user's choice.
Step5: If choice is 1 to 4 or 5:
- Read two numbers x & y

Step6: If choice is 6:
- Read one number x.

Step7: Use switch-case to perform the selected operation:
- Addition: result = x+y
- Subtraction: result = x-y
- Multiplication: result = x*y
- Division — If y≠0 result = x/y
  Else display error message.
- power: calculate x^y using loop.
- Factorial: calculate x! using loop.

Step8: Display the result
Step9: STOP

C Program:

```
# include <stdio.h>
int main() {
    int choice;
    int x, y, i;
    long int fact=1;
    int power=1;
    printf("Enter your choice:");
    scanf("%d", &choice);
    switch (choice) {
        case 1:
            printf("Enter 2 nos:");
            scanf("%d %d", &x, &y);
            printf("Result = %d", x+y);
            break;
```

```
Case 2:
        printf("Enter 2 num:");
        scanf("%d %d", &x, &y);
        printf("Result = %d", x-y);
        break;
Case 3:
        printf("Enter 2 num:");
        Scanf("%d %d", &x, &y);
        print("Result = %d", x * y);
        break;
Case 4:
        printf("Enter two num:");
        scanf("%d %d", &x, &y);
        if (y != 0)
            printf("Result = %d", x/y);
        else
            printf("Division by zero is not allowed");
        break;
Case 5;
        printf("Enter base & exponent:");
        scanf("%d %d", &x, &y);
        for (i=1; i<=y; i++)
            power = power * x;
        printf("Result = %d", power);
        break;
Case 6:
        printf("Enter a num:");
        Scanf("%d", &x);
        for(i=1; i<=x; i++)
            fact = fact * i;
        printf("Result = %Id", fact);
        break;
default:
```

```
    printf("Invalid choice");
  }
  return 0;
}
```

Input:

User selects an operation from the menu & enters required values.

Example Input:

Choice : 1

Numbers: 10 20

Output:

Example Output:

Result = 30.00

Conclusion: Thus, a simple calculator program was successfully implemented using C programming. The program demonstrates the use of switch-case statements, loops & arithmetic operations.

FAQs:

Q. What is a simple calculator program?

A. A simple calculator program is a basic computer program that performs fundamental arithmetic operations on numbers entered by the user.

2) Why is switch-case preferred in calculator programs?

A. Switch-case is preferred in calculator programs because it makes the code clear, organised, and easy to manage when there are multiple choices. Switch-case improves clarity, efficiency & maintainability.

Date / /

3. How does the program handle division by zero?

A. a) Before dividing two numbers, the program checks whether the denominator is zero.

b) If divisor is not zero, the division is performed normally.

c) If the divisor is zero, the program does not perform the division & instead display an error message.

4. How is the power operation (x^y) implemented without using math library function?

A. The power operation (x^y) is implemented without using math library functions by using a loop & repeated multiplication.

5. What is factorial & how it is calculated using loops in c?

A. The factorial of a number n is the product of all positive integers from 1 to n.

Factorial using a for loop in C.

```
#include <stdio.h>
int main() {
    int n, i;
    long long fact=1;
    printf("Enter a number:");
    Scanf("%d", &n);
    for(i=1; i<=n; i++) {
        fact = fact * i;
    }
    printf("Factorial of %d is %lld", n, fact);
    return 0;
}.
```